

	University of Cebu (UC) College of Engineering (COE) Bachelor of Science in Computer Engineering (BSCPE)		School Year:	2024-2025
			Semester/Term:	1 st
			CMO No. 87 series of 2017	

I. Course Information

Course Code:	CPE 412	LEC Units	3	Course Description: This course includes the study of the evolution of computer architecture and the factors influencing the design of hardware and software elements of computer systems. The focus is on the understanding of the design issues specifically the instruction set architecture and hardware architecture.
Course Title:	Computer Architecture and Organization	LAB Units:	1	
Pre-requisite(s):	Microprocessors	LEC Hours:	3	
Co-requisite(s):	None	LAB Hours:	3	

II. Institutional Statements of Vision, Mission, Core Values and Graduate Outcomes

UC Vision (UCV)	UC Mission (UCM)	UC Core Values (UCCV)	UC Graduate Outcomes (UCGO)
UCV1. Democratize quality education. UCV2. Be the visionary and industry leader. UCV3. Give hope and transform lives.	University of Cebu commits to: UCM1. Serve as an active catalyst in providing efficient and effective delivery of educational services; UCM2. Pursue excellence in instruction, research and community service towards social and economic development as well as environmental sustainability; and UCM3. Foster an organizational culture that nurtures employee productivity and engagement.	These are the core values that UC believes in: UCCV1. Innovation "Be the visionary and the industry leader." UCCV2. Camaraderie "Living in the spirit of harmony and approachability." UCCV3. Alignment "All activities are geared towards core values and priorities." UCCV4. Respect "Always a professional, mindful of God, university, the community and self." UCCV5. Excellence "To be great at whatever it is we do and go for the best."	UC Graduates will be able to: UCGO1. Demonstrate skills and competencies in one's educational level and field of discipline for lifelong learning; UCGO2. Express proficiency in both oral and written communication; UCGO3. Demonstrate social accountability and ethical responsibility towards the community and the environment; UCGO4. Exhibit proactive and collaborative attributes in diverse society; UCGO5. Utilize appropriate technology; and UCGO6. Manifest ethical behavior in diverse situations.

III. Departmental Statements of Vision, Mission, Goals and Core Values

COE Vision (COEV)	COE Mission (COEM)	COE Core Values (COECV)	COE Goals (COEG)
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Course Title	Date Effective:	Date Revised:	Prepared by:	Approved by:
CPE 412 – Computer Architecture and Organization	August 2025	July 7, 2025	Engr. Roland B. Fernandez, MEP	Engr. Roland B. Fernandez, MEP
			Faculty	Dean/Program Head

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The College of Engineering will be recognized as one of the leading engineering institutions in the Philippines through quality research, excellence in engineering instruction, and relevant extension services.	We commit to provide quality and affordable engineering programs that will produce engineering graduates who are responsive to the needs of the local and global community.	The Core Values of the College of Engineering are: 1.Commitment – We commit to ideals of the university, diligent performance of our duties and responsibilities, and to the education of our students. 2.Collaboration – We promote teamwork and building partnership across discipline and with the community. 3.Openness – We encourage transparency in our dealings and treat all our stakeholders fairly. 4.Excellence – We foster excellence in learning and teaching, research, and extension services. 5. Ethics – We practice professionalism, integrity, and accountability.	The College of Engineering will apply resources and pursue opportunities to achieve the following strategic goals: 1. Be the institution of first choice for high school students in Central Visayas who wish to pursue careers in engineering. 2. Produce graduates who are highly valued both local and global employers and accepted for graduate study. 3. Be a service provider of the continuing professional development for engineering and technical professionals in Central Visayas and Mindanao. 4. Establish a strong academic-industry linkage among the private corporations and government institutions.		

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IV. UC-COE Graduate Attributes

Graduate Attributes (GAs)	UCV			UCM			UCCV					UCGO						CO EV	COEM			COECV			COEG			
	1	2	3	1	2	3	1	2	3	4	5	1	2	3	4	5	6	1	1	2	3	1	2	3	1	2	3	4
GA1. Lifelong Learner																												
GA2. Effective Communicator																												
GA3. Ethically Responsible																												
GA4. Socially Collaborative																												
GA5. Team Player																												
GA6. Technically Competent																												
GA7. Entrepreneurial																												
GA8. Solution-Oriented																												

V. PEO-PILO-GA Mapping

Program Educational Objectives (PEOs)	Program Outcomes (POs)	Graduate Attributes (GAs)							
		1	2	3	4	5	6	7	8
A few years after graduation, our BSCPE graduates: PEO1. Demonstrated life-long learning through acquisition of advance degree programs, continuous engagement on personal and professional growth and development; PEO2. Shown ability to solve complex engineering problems by undertaking and involving in projects, individual or in teams, that advocates public health and safety standards, environmental preservation and rehabilitation, and community development ; and	By the time of graduation, the students of the BSCPE program shall have : PO1. An ability to apply knowledge of mathematics and science to solve complex engineering problems.								
	PO2. An ability to design and conduct experiments, as well as to analyze and interpret data.								
	PO3. An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.								
	PO4. An ability to function on multidisciplinary teams.								
	PO5. An ability to identify, formulate, and solve complex engineering problems.								
	PO6. An understanding of professional and ethical responsibility.								
	PO7. An ability to communicate effectively.								
	PO8. Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context.								
	PO9. Recognition of the need for, and an ability to engage in life-long learning.								
	PO10. Knowledge of contemporary issues.								
	PO11. An ability to use techniques, skills, and modern engineering tools necessary for engineering practice.								

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PEO3. Manifested a strong sense of professionalism through commitment to excellence, enthusiasm to learning, ethical behavior and integrity and exceptional leadership.	PO12. Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments.											

VI. PO-CILO Activation Mapping

Course Intended Learning Outcomes (CILOs)	Program Outcomes (POs)													
<i>By the end of the course, you should be able to:</i>	1	2	3	4	5	6	7	8	9	10	11	12	13	14
CILO 1. Develop a deeper understanding of computer systems design with emphasis on pipelined Intel and ARM machines;			E											
CILO 2. Know basic design principles and techniques for performance improvement.			E											
CILO 3. Use some simulation tools and appreciate the importance of simulation as the primary means of validating designs.			E											
CILO 4: Practice professional skills in design and analysis, project management, and presentation			E											

Course Outputs (appropriate rubrics may be shown below or in separate page)

CILO#	Required Output / Assessment Tasks (ATs of Learning)	Weight	Specification / Remarks
1, 2, 3	Project	20%	The project will involve a real life OS algorithm design and implementation case, which will assess how the students will make judicious choices of OS algorithm design and constructs to use to address requirements needed to solve the given problem.
1, 2, 3	Quizzes	30%	Includes chapter quizzes, long quizzes,
1, 2, 3	Laboratory Exercises	10%	
1, 2, 3	Major Examinations	30%	Prelim, Midterm, Semi-finals, Finals
1, 2, 3	Class Participation (Seatwork, Assignment, etc)	10%	

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TOTAL: 100%

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VII. Teaching and Learning Plan

CILO #	Unit Outcomes (Chapter or unit ILOs)	Topics (Course Outline)	Teaching & Learning Activities (TLAs)	Assessment Tasks (ATs for Learning)	Learning Resources / References	Time Frame
	Explain how the course relates to the program; how the program relates to the departmental and institutional graduate outcomes; how the graduate outcomes relate to UC and CCS VMGO-CV Statements.	University and College VMGO-Core Values and Graduate Outcomes Orientation	- Presentation - Discussion - Workshop	Oral Recitation	- DLP - UC Student Manual - COE Course Catalog	1 hr
		Program Educational Objectives and Program Intended Learning Outcomes Orientation				
		Course / CILO Orientation				
1	- Define Computer Organization and Computer Architecture - Differentiate Computer Organization and Computer Architecture - Illustrate the functional view of a computer system - Discuss the different sequences possible of a computer operation - Elaborate the major components of a computer system.	1. Introduction 1.1 Organization and Architecture 1.2 Structure and Function	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz	- DLP - Computer/Laptop - Simulation Software	1 hrs
1, 2	- Outline the history of computers according to different generations - Define the different techniques built in contemporary for computer processors - Examine the different ways in bridging the gap between logic and memory performance. - Outline the evolution of Intel microprocessor product line - Define the ARM architecture - Identify different ARM embedded devices	2. Computer Evolution and Performance 2.1 A Brief History of Computers The First Generation: Vacuum Tubes The Second Generation: Transistors The Third Generation: Integrated Circuits Later Generations 2.2 Designing for Performance Microprocessor Speed Performance Balance Improvements in Chip Organization and Architecture	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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	- Present an ARM System organization - Compute for Cycles per Instruction (CPI), MIPS Rate and MFLOPS Rate based on a given computer system situation.	2.3 Multicore, MICs, and GPGPUs 2.4 The Evolution of the Intel x86 Architecture 2.5 Embedded Systems and the ARM Embedded Systems ARM Evolution 2.6 Performance Assessment Clock Speed and Instructions per Second Benchmarks Amdahl's Law Little's Law				
1, 2	- List the three concepts of a Von Neuman Architecture - Define a hardwired program - Discuss the functions of a computer - Examine a program execution example. - Draw the instruction cycle state diagram - Identify the different registers utilized during program execution - Discuss the concept of Interrupts - Compare and contrast the execution of a program with no interrupts, with short wait interrupts and long wait interrupts - Enumerate the elements of Bus design - Define Point-to point interconnect - Discuss the PCI architecture	3. Top Level View of Computer Function and Interconnection 3.1 Computer Components 3.2 Computer Function Instruction Fetch and Execute Interrupts I/O Function 3.3 Interconnection Structures 3.4 Bus Interconnection Bus Structure Multiple-Bus Hierarchies Elements of Bus Design 3.5 Point-to-Point Interconnect QPI Physical Layer QPI Link Layer QPI Routing Layer QPI Protocol Layer 3.6 PCI Express PCI Physical and Logical Architecture PCIe Physical Layer	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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		PCIe Transaction Layer PCIe Data Link Layer				
1, 2, 3	- Define Cache Memory - Discuss the characteristics of a memory - Illustrate the memory hierarchy - Explain each element of cache memory design - Compute for cache addresses and size - Illustrate the memory mapping - Draw the memory utilization using LRU, LFU, FIFO and Random Replacement Algorithm - Discuss the evolution of Intel's Cache memory and ARM Cache Organization	4. Cache Memory 4.1 Computer Memory System Overview Characteristics of Memory Systems The Memory Hierarchy 4.2 Cache Memory Principles 4.3 Elements of Cache Design Cache Addresses Cache Size Mapping Function Replacement Algorithms Write Policy Line Size Number of Caches 4.4 Pentium 4 Cache Organization 4.5 ARM Cache Organization	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs
1, 2, 3	- Discuss the concept of a memory cell - Differentiate DRAM and SRAM - Illustrate the cell structures of DRAM and SRAM - Define the different types of ROM - Define Error correction code - Calculate for the Parity bits of an 8 bit data value - Identify advanced DRAM organizations	5. Internal Memory 5.1 Semiconductor Main Memory Organization DRAM and SRAM Types of ROM Chip Logic Chip Packaging Module Organization Interleaved Memory 5.2 Error Correction 5.3 Advanced DRAM Organization Synchronous DRAM	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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		Rambus DRAM DDR SDRAM Cache DRAM				
1, 2, 3	- Identify different types of external memory - Define a magnetic disk - Discuss the magnetic disk organization - Differentiate CAV and Multiple zone recording - Define the physical characteristics of a Disk System - Computer for the Rotational delay, Transfer time of disk - Define RAID - Identify and discuss the different RAID levels - Discuss the different types of Optical Disk - Discuss the process of reading or writing data of the different external memory - Discuss the properties of a Tape drive	6. External Memory 6.1 Magnetic Disk Magnetic Read and Write Mechanisms Data Organization and Formatting Physical Characteristics Disk Performance Parameters 6.2 RAID RAID Level 0 RAID Level 1 RAID Level 2 RAID Level 3 RAID Level 4 RAID Level 5 RAID Level 6 6.3 Solid State Drives Flash Memory SSD Compared to HDD SSD Organization Practical Issues 6.4 Optical Memory Compact Disk Digital Versatile Disk High-Definition Optical Disks 6.5 Magnetic Tape	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs
1, 2, 3	Explain the use of I/O modules as part of a computer organization.	7. Input/Output 7.1 External Devices Keyboard/Monitor	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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	- Understand the difference between programmed I/O and interrupt-driven I/O and discuss their relative merits. - Discuss an overview of the operation of direct memory access. - Explain the function and use of I/O channels. - Explain an overview of Thunderbolt. - Explain an overview of InfiniBand.	Disk Drive 7.2 I/O Modules Module Function I/O Module Structure 7.3 Programmed I/O Overview of Programmed I/O I/O Commands I/O Instructions 7.4 Interrupt-Driven I/O Interrupt Processing Design Issues Intel 82C59A Interrupt Controller The Intel 82C55A Programmable Peripheral Interface 7.5 Direct Memory Access Drawbacks of Programmed and Interrupt-Driven I/O DMA Function Intel 8237A DMA Controller 7.6 I/O Channels and Processors The Evolution of the I/O Function Characteristics of I/O Channels 7.7 The External Interface: Thunderbolt and InfiniBand Types of Interfaces Point-to-Point and Multipoint Configurations Thunderbolt InfiniBand				
1, 2, 3	Summarize, at a top level, the key functions of an operating system (OS).	8. Operating System Support 8.1 Operating System Overview Operating System Objectives and Functions Types of Operating Systems	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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	- Discuss the evolution of operating systems for early simple batch systems to modern complex systems. - Explain the differences among long-, medium-, and short-term scheduling. - Understand the reason for memory partitioning and explain the various techniques that are used. - Assess the relative advantages of paging and segmentation. - Define virtual memory.	8.2 Scheduling Long-Term Scheduling Medium-Term Scheduling Short-Term Scheduling 8.3 Memory Management Swapping Partitioning Paging Virtual Memory Translation Lookaside Buffer Segmentation 8.4 Pentium Memory Management Address Spaces Segmentation Paging 8.5 ARM Memory Management Memory System Organization Virtual Memory Address Translation Memory-Management Formats Access Control				
1, 2, 3	- Understand the distinction between the way in which numbers are represented (the binary format) and the algorithms used for the basic arithmetic operations. - Explain two's complement representation. - Present an overview of the techniques for doing basic arithmetic operation in two's complement notation.	9. Computer Arithmetic 9.1 The Arithmetic and Logic Unit 9.2 Integer Representation Sign-Magnitude Representation Two's Complement Representation Range Extension Fixed-Point Representation 9.3 Integer Arithmetic Negation Addition and Subtraction	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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	- Understand the use of significant, base, and exponent in the representation of floating-point numbers. - Present an overview of the IEEE 754 standard for floating-point representation. - Understand some of the key concepts related to floating-point arithmetic, including guard bits, rounding, subnormal numbers, underflow and overflow.	Multiplication Division 9.4 Floating-Point Representation Principles IEEE Standard for Binary Floating-Point Representation 9.5 Floating-Point Arithmetic Addition and Subtraction Multiplication and Division Precision Considerations IEEE Standard for Binary Floating-Point Arithmetic				
1, 2, 3	- Present an overview of essential characteristics of machine instructions. - Describe the types of operands used in typical machine instruction sets. - Present an overview of x86 and ARM data types. - Describe the types of operands supported by typical machine instruction sets. - Present an overview of x86 and ARM operation types. - Understand the differences among big endian, little endian, and bi-endian.	10. Instruction Sets: Characteristics and Function 10.1 Machine Instruction Characteristics Elements of a Machine Instruction Instruction Representation Instruction Types Number of Addresses Instruction Set Design 10.2 Types of Operands Numbers Characters Logical Data 10.3 Intel x86 and ARM Data Types x86 Data Types ARM Data Types 10.4 Types of Operations Data Transfer Arithmetic Logical Conversion	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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		Input/Output System Control Transfer of Control 10.5 Intel x86 and ARM Operation Types x86 Operation Types ARM Operation Types				
1, 2, 3	- Describe the various types of addressing modes common in instruction sets. - Present an overview of x86 and ARM addressing modes. - Summarize the issues and trade-offs involved in designing an instruction format. - Present an overview of x86 and ARM instruction formats. - Understand the distinction between machine language and assembly language.	11. Instructions Sets: Addressing Modes and Format 11.1 Addressing Immediate Addressing Direct Addressing Indirect Addressing Register Addressing Register Indirect Addressing Displacement Addressing Stack Addressing 11.2 x86 and ARM Addressing Modes x86 Addressing Modes ARM Addressing Modes 11.3 Instruction Formats Instruction Length Allocation of Bits Variable-Length Instructions 11.4 x86 and ARM Instruction Formats x86 Instruction Formats ARM Instruction Formats	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs
1, 2, 3, 4	- Distinguish between user-visible and control/status registers, and discuss the purposes of registers in each category. - Summarize the instruction cycle.	12. Processor Structure and Function 12.1 Processor Organization 12.2 Register Organization User-Visible Registers Control and Status Registers	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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	- Discuss the principle behind instruction pipelining and how it works in practice. - Compare and contrast the various forms of pipeline hazards. - Present an overview of the x86 processor structure. - Present an overview of the ARM processor structure.	Example Microprocessor Register Organizations 12.3 Instruction Cycle The Indirect Cycle Data Flow 12.4 Instruction Pipelining Pipelining Strategy Pipeline Performance Pipeline Hazards Dealing with Branches Intel 80486 Pipelining 12.5 The x86 Processor Family Register Organization Interrupt Processing 12.6 The ARM Processor Processor Organization Processor Modes Register Organization Interrupt Processing				
1, 2, 3, 4	- Provide an overview research results on instruction execution characteristics that motivated the development of the RISC approach. - Summarize the key characteristics of RISC machines. - Understand the design and performance implications of using a large register file. - Understand the use of compiler-based register optimization to improve performance.	Reduced Instruction Set Computers 13.1 Instruction Execution Characteristics Operations Operands Procedure Calls Implications 13.2 The Use of a Large Register File Register Windows Global Variables Large Register File versus Cache 13.3 Compiler-Based Register Optimization 13.4 Reduced Instruction Set Architecture	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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	- Discuss the implication of a RISC architecture for pipeline design and performance. - List and explain key approaches to pipeline optimization on a RISC machine.	Why CISC Characteristics of Reduced Instruction Set Architectures CISC versus RISC Characteristics 13.5 RISC Pipelining Pipelining with Regular Instructions Optimization of Pipelining 13.6 MIPS R4000 Instruction Set Instruction Pipeline 13.7 SPARC SPARC Register Set Instruction Set Instruction Format 13.8 RISC versus CISC Controversy				
1, 2, 3	- Explain the difference between superscalar and superpipelined approaches. - Define instruction-level parallelism. - Discuss dependencies and resource conflicts as limitations to instruction level parallelism - Present an overview of the design issues involved in instruction-level parallelism. - Compare and contrast techniques of improving pipeline performance in RISC machines and superscalar machines.	14. Instruction Level Parallelism and Superscalar Parallelism 14.1 Overview Superscalar versus Superpipelined Limitations 14.2 Design Issues Instruction-Level Parallelism and Machine Parallelism Instruction Issue Policy Register Renaming Machine Parallelism Branch Prediction Superscalar Execution Superscalar Implementation	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs
1, 2, 3	Discuss the Micro operations of the Control Unit	15. Control Unit Operation 15.1 Micro-Operations	- Lecture - Discussion	- Seat-work - Quiz	- DLP - Computer/Laptop	3 hrs

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	- Identify the flow of the instruction cycle. - Determine the functions that the control unit must perform to cause the micro-operations to be performed. - Familiarize the block diagram of the Control Unit - Provide a typical list of the inputs and outputs of a control unit. - List three types of control signals. - Briefly explain what is meant by a hardwired implementation of a control unit	The Fetch Cycle The Indirect Cycle The Interrupt Cycle The Execute Cycle The Instruction Cycle 15.2 Control of the Processor Functional Requirements Control Signals A Control Signals Example Internal Processor Organization The Intel 8085 15.3 Hardwired Implementation Control Unit Inputs Control Unit Logic	Hands-on Laboratory	Laboratory Exercises	Simulation Software	
1, 2, 3	- Discuss the principles of microprogrammed control - Enumerate the advantages and disadvantages of microprogrammed - Identify the design consideration of microinstruction sequencing - Outline the sequence of microinstruction execution	16. Microprogrammed Control 16.1 Basic Concepts Microinstructions Microprogrammed Control Unit Wilkes Control Advantages and Disadvantages 16.2 Microinstruction Sequencing Design Considerations Sequencing Techniques Address Generation LSI-11 Microinstruction Sequencing 16.3 Microinstruction Execution A Taxonomy of Microinstructions Microinstruction Encoding LSI-11 Microinstruction Execution IBM 3033 Microinstruction Execution	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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	- Discuss the principles of multicore computers - Enumerate and discuss the issues of multicore technologies in both hardware and software performance - Explain the multicore organization - Illustrate the x86 organization.	17. Multicore Computer 17.1 Hardware performance issues 17.2 Software performance issues 17.3 Multicore organization 17.4 Intel x86 organization	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs
	- Explain the advantages and disadvantages of different distributed systems architecture. - Discuss client-server and distributed object architecture. - Discuss the underlying principles of the CORBA standards. - Identify the new models in distributed computing.	18. Distributed Systems Architecture	- Lecture - Discussion - Hands-on Laboratory	- Seat-work - Quiz - Laboratory Exercises	- DLP - Computer/Laptop - Simulation Software	3 hrs

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Textbook(s) and References	
Textbook (APA Format):	References (APA Format including citations):
William Stallings. Computer Organization and Architecture: Designing for Performance. 9 th Ed. Pearson Education, Inc.	Henessy J and Patterson D. Computer Architecture: A Quantitative Approarch. 5 th Edition Yadin A., (2016) Computer Systems Architecture. CRC Press. Taylor & Francis Group Null L and Loburr J. (2003) The Essentials of Computer Organization and Architecture. Jones and Barlett Publishers Stallings, W (2013) Computer Organization and Architecture: International Edition. 8 th Edition, Pearson Education Limited UC. (2011). <i>Students Manual 2015</i> . Cebu: Rex BookStore.

VIII. Other Information (Classroom Rules and Policies/ Teacher Information, etc.)

Classroom Rules / Policies	Teacher Information
1. Cheating is not allowed. 2. Textbook is required. 3. School policy for required attendance/tardiness is followed (maximum of 9 absences for MWF classes and 7 absences for TTH classes) 4. Three (3) late is equivalent to 1 absent. 5. Laboratory exercises and Machine Project must be submitted on time. Late submission is not accepted. 6.	Engr. Roland B. Fernandez, MEP

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CPE 412 – Computer Architecture and Organization	August 2025	July 7, 2025	Engr. Roland B. Fernandez, MEP	Engr. Roland B. Fernandez, MEP
			Faculty	Dean/Program Head